KAMALESH JAYAPANDIARAJ ARUMUGAM

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OSCP | Cloud & Application Security Engineer | Policy-as-Code · Cloud IAM · API Security · DevSecOps

SUMMARY

Cloud security engineer who builds preventive controls and the tooling that proves them: policy-as-code across AWS, GCP and OCI, IAM least privilege, and guardrail validation enforced pre-merge in CI. Offensive background (OSCP, CPTC) used to demonstrate exploitability in production and then eliminate the class, not file the finding. Owns cross-cloud enforcement mapping inside a Fortune 100 policy-as-code program in PCI-scoped environments.

EXPERIENCE

American Express (via Tata Consultancy Services) **Phoenix, AZ, USA**

Cloud Security Engineer *Sept 2026 – Present*

- OCI **security owner** within a **multi-cloud policy-as-code program** (AWS / GCP / OCI) at a Fortune 100 financial institution; author and maintain **50+ native IAM** and resource policies across **~20 compartments** in **PCI-scoped environments**.
- Authored and amended enforcement policy across **AWS service control policies, GCP organization policies, and OCI-native equivalents**, owning the **control-mapping layer** that holds 10+ control families consistent across three clouds.
- Designed and shipped a **Go service that evaluates policy guardrails against Terraform plans** pre-merge in CI, versioned, test-covered, and adopted as a required check across 3 repositories, catching 20+ policy regressions before merge and cutting review turnaround from ~3 days to ~1 day; designated reviewer on all OCI policy pull requests.
- Identified **5+ IAM privilege-escalation paths** in production tenancies through **policy-graph analysis**, demonstrated exploitability in an isolated compartment under written authorization, then authored **preventive policy and an audit-log detection** that eliminated the class across ~20 compartments.
- Extended policy enforcement to **container workloads**, scoping workload identities to **least-privilege cloud access, enforcing image provenance, and gating deployments** through admission control across 6 clusters and 80 namespaces, covering 400+ workloads.
- Led **threat modelling for network connectivity and data-flow designs** in PCI-scoped environments, identifying trust-boundary gaps that changed the approved architecture and **eliminated public egress paths** from 10+ workloads.

Keck Medicine of USC **Los Angeles, CA, USA**

Cloud Security Engineer Intern (Part-Time) *Oct 2024 – May 2026*

- Automated **compliance enforcement** across **100+ AWS resources**, codifying CIS and HIPAA requirements as **AWS Config rules with Python Lambda remediation** wired into Security Hub: cut open findings ~65% and removed ~10 hrs/week of manual review.
- Built the Keck Cloud Portal (Terraform, Lambda, policy-as-code) as a paved-road **infrastructure provisioning** path with security baselines enforced at deploy time, cutting environment onboarding from 2–3 days to under 1 day.
- Discovered an **IDOR / BOLA authorization** flaw exposing ~1,000 records; **demonstrated exploitability, drove the object-level authorization fix, and added regression coverage** to prevent recurrence of the pattern.

Oracle Corporation **Bangalore, KA, India**

Associate Cloud Engineer, JAPAC *Sept 2022 – July 2024*

- Led **an IAM entitlement audit** across 10+ cloud tenancies, rebuilding role and policy structures around least privilege and eliminating 90% of standing privilege.
- Mapped **detective controls** to the CIS Benchmark and tuned automated **posture-management recipes**, raising benchmark compliance from 58% to 96%.
- **Hardened 20+ production workloads** with network security groups, subnet segmentation, and centralized key management (Vault / KMS) under a defense-in-depth baseline.

PROJECTS

- **LLM & Agentic Security Red-Team Test Suite** — (Github) Python test-suite covering **tool-call injection, indirect/RAG injection, excessive agency, cross-agent trust, and memory poisoning**, mapped to the OWASP LLM Top 10. Built around capability-scoped tool authorization; constraining what an agent identity may invoke rather than filtering what it says.
- **Authorization Testing Framework for REST APIs** — (Github) Automated Broken object-level authorization / IDOR and **privilege-boundary testing** across authenticated roles; generates per-endpoint authorization matrices and **flags object-level access gaps in CI.**

EDUCATION

University of Southern California - M.S. Cybersecurity Engineering · GPA 3.9 / 4.0 · Los Angeles, CA **Aug 2024 – May 2026**

CERTIFICATIONS & AWARDS

- **Certifications:** OSCP | OCI Architect Professional | Microsoft SC-900 | CompTIA Security+ | AWS Security Specialty (In Progress)
- **Awards:** 25/143: Amazon AppSec CTF (NAMER) | 2nd Place: Tenable × CRACCON 2025 Cloud & CNAPP CTF | Top 10 : WRCCDC 2026 Offensive Security Team, CPTC 2025 (USC) | 2nd Runner-Up: USC CTF, Fall 2024

TECHNICAL SKILLS

Cloud Security: AWS (IAM, Config, Security Hub, GuardDuty, Lambda, KMS, S3) · OCI (IAM, Policies, Cloud Guard, VCN, Vault/KMS) · Azure AD / Entra ID · GCP · Kubernetes · Container Security

Policy-as-Code & IaC: HashiCorp Sentinel · Terraform · AWS Config Rules · CI/CD policy gates · GitHub Actions · OPA/Rego · Conftest

Application Security: API security · OWASP API Top 10 · AuthN / AuthZ · BOLA / IDOR · OWASP Top 10 & ASVS · threat modeling (STRIDE) · secure SDLC · SAST / DAST · LLM & AI security

Languages, Tools & Frameworks: Python · Go · Bash · SQL · YAML · Burp Suite · OWASP ZAP · Semgrep · Trivy · Nmap · CIS Benchmarks · PCI DSS · HIPAA